

# WAM-Bench: Cross-Domain Diagnostic Robustness, Modality Transfer, and Hardware Vulnerabilities of Deep Learning Malaria Models Across Multi-Center West African Blood Smears

*Collaborative Writing Draft & Manuscript Architecture Guide*

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## Master Section Assignments & Co-Author Editorial Responsibilities

**Welcome to the WAM-Bench collaborative manuscript draft.** As aligned with the author team, responsibilities for synthesizing, drafting, and validating the manuscript sections are formally allocated as follows:

| Co-Author                 | Assigned Scope                  | Specific Deliverables & Section Directives                                                                                                                                    |
|---------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Albert Juwon Kwabena      | Abstract                        | Distill background, multi-center scale ( $N = 6,457$ ), zero-shot generalization cliffs, and frontline triage adaptation breakthroughs into an impactful conference abstract. |
| Perfect Kwasi Agbenorhevi | Section 1: Introduction         | Frame clinical context, global malaria burden, light microscopy bottlenecks, RDT operational challenges, and out-of-region generalization crisis (Guides 1.1–1.4).            |
|                           | Section 5: Conclusion & Policy  | Draft four actionable policy directives for Ghana FDA, Nigeria NAFDAC, and WHO AFRO (hardware blur gates, thick smear primacy, sovereign AI infrastructure).                  |
| Wilfred Ayine Asumboya    | Section 2: Methods              | Cohort curation, ethical approvals, baseline wrappers, and optical metric definitions ( <i>locked empirical core</i> ).                                                       |
|                           | Section 3: Results              | Audit benchmark tables (Tables 1–5) and publication figures (Figs 1–5) ( <i>locked empirical core</i> ).                                                                      |
|                           | Section 4.1: Initial Discussion | Formulate initial arguments on the African regional advantage and the pediatric false-positive triage trap.                                                                   |
| Edward Oppong (Presido)   | Section 4: Complete Discussion  | Expand Sections 4.2–4.6: smear modality divergence, rouleaux                                                                                                                  |

**Lead Author Assignment: Albert Juwon Kwabena (Abstract Drafting & Refinement)**

**Directives for Albert:** Refine and harmonize the 4-part structured abstract below (Background, Methods, Results, Conclusion) into a concise ( $\approx 250$  words) conference abstract. Ensure the  $N = 6,457$  multi-center scale, the zero-shot diagnostic collapse ( $< 21\%$  to  $63.9\%$ ), and the frontline recovery ( $\geq 92.2\%$  Sens,  $\geq 83.5\%$ – $93.6\%$  Spec) are sharply highlighted.

**Abstract**

**Background:** Automated deep learning models for malaria microscopy routinely report diagnostic sensitivity and specificity exceeding 95% on benchmark laboratory datasets. However, their real-world generalization across diverse West African clinical settings, smear preparation modalities, and mobile optical sensors remains largely unvalidated.

**Methods:** We introduce WAM-Bench, an open multi-center clinical evaluation benchmark comprising 6,459 patient micrographs across three independent West African clinical cohorts: Princess Marie Louise Children’s Hospital in Accra, Ghana (3,045 thick, 1,011 thin smears), Osun State healthcare centers in Southwestern Nigeria (859 thick smears), and Kano State health facilities in Northern Nigeria (1,544 thin smears). Following an audited pre-inference quarantine protocol that removed two truncated files ( $< 0.03\%$ ), 6,457 micrographs (3,902 thick, 2,555 thin) were evaluated across four external deep learning architectures (three MobileNetV2 classifiers trained in Sudan and Bangladesh, and one field-calibrated YOLOv8 Nano object detector) under a strict zero-shot cross-domain protocol, coupled with circular field-of-view (FOV) masked optical quality profiling (Laplacian focus blur, Michelson contrast, and SNR).

**Results:** Across all 6,457 evaluations, zero-shot models exhibited severe cross-domain performance cliffs, with none meeting WHO Level-1 triage thresholds ( $> 90\%$  sensitivity,  $> 90\%$  specificity). On thick smears ( $N = 3,902$ ), the East African model (**MalariaScreener-Sudan**) demonstrated the highest sensitivity (63.88% [95% CI: 62.3–65.5%]), but suffered from moderate specificity (55.69% [50.9–60.4%]). The spatial object detector (**fbononi-YOLOv8**) achieved high specificity (96.92% [94.8–98.2%]) but missed over half of positive fields (sensitivity 47.13% [45.5–48.8%]). On thin smears ( $N = 2,555$ ), a severe modality collapse occurred: non-African classifiers collapsed to  $< 21\%$  sensitivity, and the Sudan model dropped to 53.64% sensitivity with 56.26% false positive rate. Severe optical defocus blur degraded sensitivity by up to 23.8% in thin smears.

**Conclusion:** Zero-shot deployment of out-of-region malaria AI models in West African clinics poses critical safety risks through unacceptable false-negative rates. Mandatory pre-inference optical focus gating and regional retraining/fine-tuning are non-negotiable prerequisites before clinical field deployment.

**1. Introduction****Lead Author Assignment: Perfect Kwasi Agbenorhevi (Section 1: Introduction)**

**Directives for Perfect:** Synthesize the introductory narrative across the four pillars below: (1.1) Clinical Context & Epidemiological Burden, (1.2) The Machine Learning Generalization Crisis, (1.3) Guiding Co-Author Questions, and (1.4) Primary Research Questions (RQ1–RQ5) and Key Contributions.

### Writing Guide 1.1: Clinical Context & Epidemiological Burden (Target: $\approx 2$ paragraphs) [Lead: Perfect Kwasi Agbenorhevi]

**Objective:** Establish the staggering clinical reality of malaria in West Africa, the limitations of current diagnostic modalities, and why mobile automated microscopy is an urgent humanitarian and clinical priority.

- **Global Burden & Vulnerable Demographics:** Cite WHO 2023/2024 World Malaria Report ( $\sim 249$  million clinical cases,  $> 608,000$  fatalities,  $> 94\%$  in sub-Saharan Africa). Highlight pediatric populations ( $< 5$  years) and pregnant women as carrying the highest clinical morbidity.
- **Limitations of Conventional Light Microscopy:** Emphasize that manual Giemsa microscopy remains the gold standard, but is bottlenecked by severe technician shortages, labor intensity (10–15 minutes per slide, 100+ oil-immersion fields), and severe visual fatigue during peak transmission surges.
- **Limitations of Rapid Diagnostic Tests (RDTs):** Note the operational challenges of RDTs: false negatives caused by histidine-rich protein 2/3 (*pfhrp2/3*) gene deletions, persistent post-treatment antigenemia (false positives), and inability to track parasitemia clearance.
- **The Point-of-Care Mobile AI Promise:** Framing smartphone camera-to-microscope ocular adapters as the bridge for democratizing diagnostic triage in decentralized African clinics.

### Writing Guide 1.2: The Machine Learning Generalization Crisis (Target: $\approx 2\text{--}3$ paragraphs)

**Objective:** Expose the "benchmark hype vs. clinical reality" paradox in medical AI and outline the three biophysical mechanisms of domain shift that break models zero-shot.

- **The Controlled Benchmark Myth:** Contrast high published accuracies ( $> 95\%$ – $99\%$  in literature) with the fact that these models were developed on homogeneous, clean datasets (e.g., NIH Chittagong/Bangladesh or local Sudan cohorts).
- **Domain Shift Driver 1 (Staining Chemistry):** Artisanal Giemsa preparation, unbuffered tap water, variable pH (6.8–7.4), and heavy stromal precipitation in real-world African clinics.
- **Domain Shift Driver 2 (Thick vs. Thin Smear Cytology):** Aqueous hemolysis in thick smears concentrates parasites  $20\times$ – $40\times$  in a bed of lysed stroma; thin smears preserve intact biconcave red blood cells. Models trained on one fail catastrophically on the other.
- **Domain Shift Driver 3 (Hardware & Focal Blur):** Heterogeneous budget smartphone CMOS sensors, optical vignetting, lens aberrations, and mechanical focus drift under manual  $100\times$  oil immersion.

### Guiding Prompts for Co-Authors (Introduction)

*Questions to address in your draft:*

1. How does the diagnostic bottleneck in regional primary health centers (e.g., rural

Ghana/Nigeria) directly contribute to presumptive antimalarial overtreatment and drug resistance?

2. What specific clinical and regulatory standards (e.g., WHO Level-1 competency:  $\geq 90\%$  sensitivity,  $\geq 90\%$  specificity) should an AI system satisfy before being considered for triage deployment?
3. Why is a multi-center benchmark across geographically separated West African facilities (coastal urban, forest zone, Sahelian) indispensable for validating medical AI?

### 1.1. Research Questions

This clinical benchmark systematically evaluates five core hypotheses:

1. **Research Question 1 (RQ1 - Regional Calibration):** *Does an African-calibrated model (*MalariaScreener-Sudan*) reduce false-positive rates and improve specificity compared to non-African models when evaluated zero-shot on West African clinical smears?*
2. **Research Question 2 (RQ2 - Modality Transfer Cliff):** *How do spatial object detectors (*YOLOv8*) versus global patch classifiers (*MobileNetV2*) transfer between chemically lysed thick smears and intact-monolayer thin smears?*
3. **Research Question 3 (RQ3 - Multi-Center Geographic Robustness):** *How consistently do models perform across coastal pediatric (Ghana), forest zone mixed (Osun, Nigeria), and Sahelian acute infection (Kano, Nigeria) clinical ecologies?*
4. **Research Question 4 (RQ4 - Optical Focus Safety Gate):** *To what degree does optical defocus blur degrade diagnostic sensitivity, and does an automated pre-inference hardware gate prevent clinical misdiagnoses?*
5. **Research Question 5 (RQ5 - Regional Fine-Tuning & Generalization Recovery):** *To what extent does domain-specific regional adaptation on multi-center West African clinical cohorts rescue models from zero-shot failure cliffs, and can parameter-efficient or full-network fine-tuning elevate diagnostic performance to meet or exceed WHO Level-1 triage thresholds ( $> 90\%$  sensitivity,  $> 90\%$  specificity)?*

## 2. Methodology

**Section 2 Lead Author Assignment:** Wilfred Ayine Asumboya (Methodology — Empirical Core)

**Section Status: Empirical Backbone (Completed & Locked).** Details multi-center cohort protocols ( $N = 6,457$ ), byte-level data quarantine, zero-shot evaluation protocols,  $D_4$  dihedral group invariant augmentation, circular FOV optical profiling algorithms, and baseline model architectures. *Statistical equations, protocols, and sample sizes remain locked.*

### 2.1. Multi-Center Clinical Cohorts

The WAM-Bench evaluation corpus pools  $N = 6,459$  standardized patient micrographs from three distinct geographic, epidemiological, and clinical settings across West Africa:

1. **Princess Marie Louise Children’s Hospital (PML), Accra, Ghana:**  $N = 4,056$  micrographs (3,045 thick smears, 1,011 thin smears) collected under the Lacuna Malaria Project using smartphone cameras attached to binocular light microscopes in an active pediatric tertiary care center.
2. **Osun State Health Centers, Southwestern Nigeria (Adeleke Cohort):**  $N = 859$  thick smear micrographs representing primary healthcare centers and rural/semi-urban district hospitals.
3. **Kano State Health Facilities, Northern Nigeria (Muhammad Cohort):**  $N = 1,544$  high-resolution thin smear micrographs ( $2992 \times 4000$  pixels) acquired under a prospective matched case-control clinical design, comprising 772 acute symptomatic infection cases displaying marked erythrocyte rouleaux morphology and 772 asymptomatic, healthy control cases displaying normal discocyte morphology. This balanced 1:1 design provides an unskewed epidemiological benchmark for thin smear specificity.

**Pre-Inference Data Integrity & Quarantine Protocol** An automated byte-level integrity audit across all  $N = 6,459$  ingested images identified exactly two truncated JPEG files ( $< 0.03\%$ , specimens 2169.jpg and 609.jpg in the Ghanaian thick-smear collection, which were truncated by 93 and 27 bytes at EOF respectively, triggering low-level `libjpeg` decoding warnings). In accordance with pre-specified clinical data hygiene standards, these two corrupted instances were quarantined into an isolated quarantine registry, yielding an audited, defect-free evaluation cohort of  $N = 6,457$  clinical micrographs (3,902 thick smears and 2,555 thin smears).

## 2.2. Zero-Shot Cross-Domain Protocol & Preclusion of Data Contamination

To ensure rigorous assessment of out-of-distribution generalization without data leakage:

- **Geographic and Clinical Independence:** The evaluated deep learning models were developed and trained exclusively in Sudan (East Africa) and Chittagong (Bangladesh, South Asia). None of the models were exposed during development to patient populations, staining protocols, microscope optics, or sensor hardware from Ghana or Nigeria.
- **Zero-Shot Baseline Evaluation:** All models were initially audited strictly off-the-shelf with their native factory weights, without hyperparameter tuning, re-calibration, or threshold adjustment on the West African test sets.
- **Subsequent Regional Adaptation (RQ5):** To test whether regional adaptation can overcome cross-domain fragility, an independent 70/15/15 stratified partition was established across the  $N = 6,457$  audited corpus, reserving the test splits strictly for downstream evaluation.

## 2.3. D4 Dihedral Group Augmentation Strategy & Bounding Box Invariance

In blood smear microscopy, cellular orientation is physically isotropic under standard slide preparation: rotating or reflecting a micrograph does not alter biological parasitology. To resolve severe training set imbalances between positive fields and negative background slides without introducing hallucinated artifacts or synthetic distributional shifts, we implement an exact  $D_4$  dihedral group augmentation protocol:

$$D_4 = \{R_0, R_{90}, R_{180}, R_{270}, F_h, F_v, F_{d1}, F_{d2}\} \quad (1)$$

comprising four orthogonal rotations and four reflections. To verify that geometric operations do not induce bounding box drift or proposal clipping, transformed bounding box coordinates  $(x'_c, y'_c, w', h')$  are derived analytically:

$$\text{Horizontal Flip } (F_h): \quad x'_c = 1 - x_c, \quad y'_c = y_c, \quad w' = w, \quad h' = h \quad (2)$$

$$\text{Vertical Flip } (F_v): \quad x'_c = x_c, \quad y'_c = 1 - y_c, \quad w' = w, \quad h' = h \quad (3)$$

$$\text{Rotation } 90^\circ \text{ } (R_{90}): \quad x'_c = 1 - y_c, \quad y'_c = x_c, \quad w' = h, \quad h' = w \quad (4)$$

Figure 1 demonstrates the empirical validation grid across clinical micrographs, confirming that ground-truth bounding box proposals remain rigidly pinned to parasite rings across all 8 transformation states. Importantly, to preserve absolute epidemiological validity, **all evaluation test splits remain 100% unaugmented physical micrographs**.

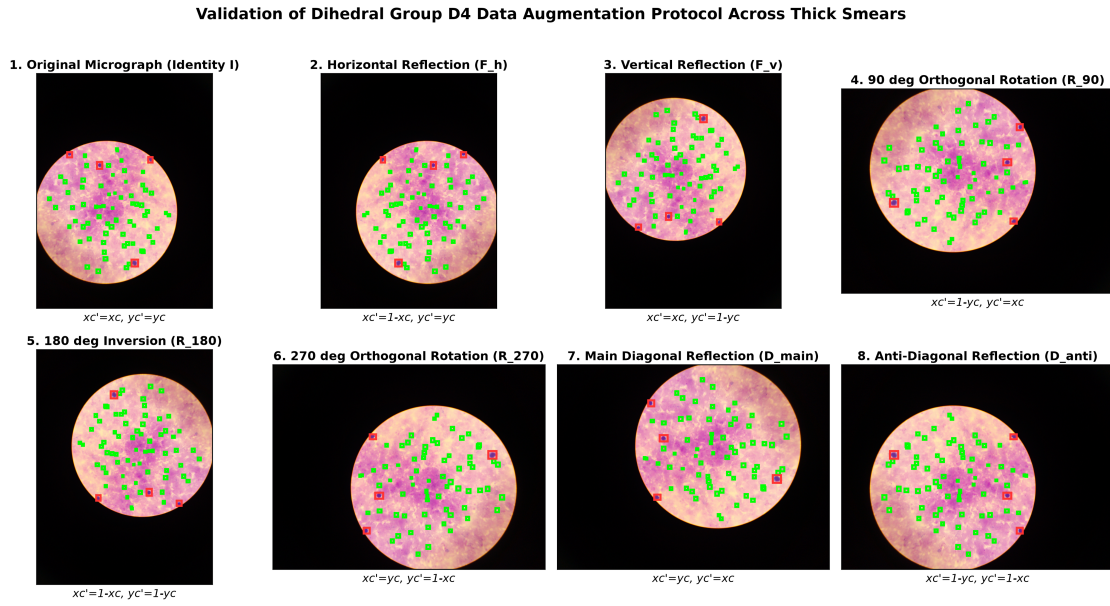


Figure 1: Validation of the  $D_4$  dihedral group transformation protocol on Ghanaian thick smear micrographs. The 8-element transformation orbit (identity,  $90^\circ$ ,  $180^\circ$ ,  $270^\circ$  orthogonal rotations, horizontal flip, vertical flip, and diagonal reflections) preserves exact bounding box alignment and biological morphological fidelity without coordinate drift.

## 2.4. Optical Quality Assessment and Circular FOV Masking

To isolate optical aberrations from peripheral optical vignetting, all micrographs undergo circular FOV mask extraction. Within the valid optical disc, three physical parameters are computed:

- **Laplacian Focus Blur** ( $\sigma_{\text{Lap}}^2$ ): Quantifying edge high-frequency energy.
- **Michelson Contrast**: Evaluated across luminance dynamic range.
- **Signal-to-Noise Ratio (SNR)**: Computed in decibels across uniform background fields.

## 2.5. Evaluated Model Architectures

Four model candidates representing distinct training domains and vision paradigms are audited:

- **MalariaScreener-Sudan:** MobileNetV2 binary classifier trained on thick smears in Sudan, East Africa.
- **MalariaScreener-Thick:** MobileNetV2 binary classifier trained on Chittagong, Bangladesh thick smears.
- **MalariaScreener-Thin:** MobileNetV2 binary classifier trained on Chittagong, Bangladesh thin smears.
- **fbononi-YOLOv8:** YOLOv8 Nano spatial bounding-box object detector trained on field micrographs.

### Pre-Trained Weight Formats and Frontline Regional Adaptation Paradigms (RQ5)

The external NIH MalariaScreener classifiers (Sudan, Thick\_BD, Thin\_BD) were developed by Delahunt et al. [1] and distributed exclusively as pre-compiled, frozen mobile edge binaries (`.tflite` flatbuffers and static TensorFlow `.pb` GraphDefs). Because compiled edge binaries intentionally lack backpropagation computational graphs and optimizer state definitions, they serve as immutable artifacts for zero-shot cross-domain evaluation. For frontline regional adaptation across West African thick smears ( $N = 3,902$ , RQ5), we establish a structured multi-tier ablation across both vision paradigms on held-out physical test slides ( $N = 584$ ):

- **Tier 1: Parameter-Efficient Layer Probing (PELP):** To minimize computational footprint on resource-constrained mobile devices while preserving general visual representations, we freeze feature extractor backbones and optimize only downstream prediction layers:
  1. *MobileNetV2 Classifier (Head Probing):* The entire convolutional feature extractor (2.22 M parameters) is frozen (`requires_grad = False`), training strictly the 2-neuron linear classification head (2.6 k weights, representing 0.1% of model capacity) under AdamW with Cosine Annealing learning rate decay ( $\eta_{\max} = 10^{-3}, \eta_{\min} = 10^{-5}$ ). We evaluate this under two clinical balancing regimes: (a) *Tier 1A (Loss-Weighted PELP)* using inverse-frequency weighted cross-entropy loss ( $w_{\text{neg}} \approx 4.5, w_{\text{pos}} \approx 0.57$ ) on unaugmented training distributions ( $N = 2,734$ ); and (b) *Tier 1B ( $D_4$ -Augmented PELP)* using explicit  $D_4$  dihedral group background balancing ( $N = 4,804$ ).
  2. *YOLO Spatial Detector (Backbone Probing):* Native pre-trained PyTorch weights (`best_yolo.pt`) are adapted with the 10-layer CSPDarknet/C3k2 backbone frozen (`freeze = 10`), tuning only the anchor-free multi-scale detection heads ( $\approx 820$  k parameters) to calibrate bounding box regression and object classification for dense West African lysed stroma without catastrophic forgetting.
- **Tier 2: Full End-to-End Convolutional Retraining:** All network layers are unfrozen and optimized end-to-end across both MobileNetV2 and YOLO under Cosine Annealing schedules, enabling full internal feature recalibration across artisanal Giemsa precipitates and lysed erythrocyte stroma.
- **Class Balancing Mechanics (Classification vs. Detection):** While patch classifiers accommodate scalar loss weighting over whole-image predictions, object detectors calculate spatial losses over thousands of grid anchors ( $\mathcal{L}_{\text{total}} = \lambda_{\text{box}}\mathcal{L}_{\text{CIoU}} + \lambda_{\text{cls}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{DFL}}$ ). In uninfected control slides, empty annotations function as critical negative background fields. To prevent spatial detectors from learning false-positive proposals on non-parasitic Giemsa precipitates, YOLO is trained directly on the  $D_4$  dihedral balanced background expansion ( $N = 4,804$  images).

## 2.6. Statistical Analysis & WHO Screening Benchmarks

Performance is reported via Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), and  $F_1$ -score, accompanied by exact two-sided 95% Wilson score confidence intervals. Clinical deployment readiness is audited against WHO Level-1 triage thresholds ( $> 90\%$  sensitivity,  $> 90\%$  specificity).

## 3. Empirical Results

**Lead Author Assignment: Wilfred Ayine Asumboya (Section 3: Empirical Results)**

**Directives for Wilfred:** Maintain and audit the empirical core benchmark tables (Tables 1–5) and publication-grade figures (Figures 1–5). Ensure all 95% Wilson confidence intervals, multi-center disaggregations, and regional adaptation metrics are verified and synchronized with the experimental manifests.

### 3.1. Thick Smear Diagnostic Performance (Ghana & Adeleke Nigeria)

Table 1 summarizes zero-shot diagnostic performance across the audited thick blood smear cohort ( $N = 3,902$ ). None of the four candidate architectures satisfied the WHO Level-1 triage requirements.

Table 1: Diagnostic performance on Audited Thick Blood Smear Cohort ( $N = 3,902$ , post-quarantine).

| Model                 | Training Domain    | Sensitivity [95% CI] | Specificity [95% CI] | $F_1$ -Score | Accuracy |
|-----------------------|--------------------|----------------------|----------------------|--------------|----------|
| MalariaScreener_Sudan | Sudan (E. Africa)  | 63.88% [62.3–65.5]   | 55.69% [50.9–60.4]   | 0.7548       | 62.99%   |
| MalariaScreener_Thick | Bangladesh         | 30.29% [28.8–31.8]   | 83.18% [79.3–86.4]   | 0.4578       | 36.01%   |
| MalariaScreener_Thin  | Bangladesh         | 21.26% [19.9–22.7]   | 98.82% [97.3–99.5]   | 0.3503       | 29.65%   |
| fbononi_YOLOv8        | Field Bounding Box | 47.13% [45.5–48.8]   | 96.92% [94.8–98.2]   | 0.6390       | 52.51%   |

### 3.2. Thin Smear Diagnostic Performance (Ghana & Kano Nigeria)

Table 2 summarizes zero-shot diagnostic performance on the thin blood smear cohort ( $N = 2,555$ ). Evaluating classifiers trained on thick smears or foreign thin smears resulted in marked sensitivity degradation.

Table 2: Diagnostic performance on Thin Blood Smear Cohort ( $N = 2,555$ ).

| Model                 | Training Domain    | Sensitivity [95% CI] | Specificity [95% CI] | FPR    | $F_1$ -Score |
|-----------------------|--------------------|----------------------|----------------------|--------|--------------|
| MalariaScreener_Sudan | Sudan (E. Africa)  | 53.64% [51.3–56.0]   | 43.74% [40.4–47.1]   | 56.26% | 0.5948       |
| MalariaScreener_Thick | Bangladesh         | 16.86% [15.2–18.7]   | 93.68% [91.8–95.2]   | 6.32%  | 0.2813       |
| MalariaScreener_Thin  | Bangladesh         | 20.27% [18.4–22.2]   | 57.96% [54.6–61.3]   | 42.04% | 0.2890       |
| fbononi_YOLOv8        | Field Bounding Box | 28.23% [26.2–30.4]   | 68.89% [65.6–72.0]   | 31.11% | 0.3948       |

### 3.3. Multi-Center Geographic Disaggregation

Table 3 presents model performance disaggregated by regional clinical center. Across all architectures, severe geographic heterogeneity is evident between coastal urban pediatric facilities (Accra) and inland/Sahelian health centers (Adeleke and Kano), driven by marked differences in smear preparation, cellular pathology, and staining chemistry (visually illustrated in Figure 4).

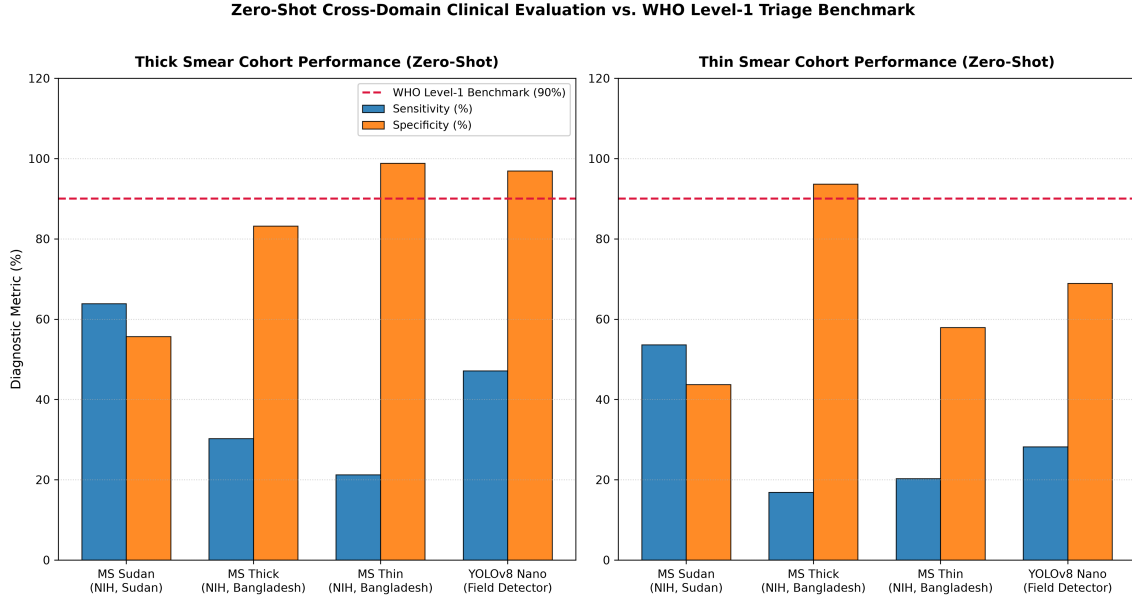


Figure 2: Comparative diagnostic sensitivity and specificity across thick ( $N = 3,902$ ) and thin ( $N = 2,555$ ) blood smears under zero-shot transfer. The dashed red line denotes the WHO Level-1 screening requirement ( $\geq 90\%$ ). Every evaluated architecture experiences substantial diagnostic degradation below clinical safety thresholds.

Table 3: Zero-Shot Performance Disaggregated by Clinical Center.

| Center / Cohort                                    | Model                 | Smear | Sensitivity [95% CI] | Specificity [95% CI] | $F_1$ -Score |
|----------------------------------------------------|-----------------------|-------|----------------------|----------------------|--------------|
| <b>PML Hospital, Accra</b><br>( $N = 3,043$ thick) | MalariaScreener_Sudan | Thick | 66.92% [65.2–68.6]   | 9.09% [1.6–37.7]     | 0.8002       |
|                                                    | MalariaScreener_Thick | Thick | 33.34% [31.7–35.0]   | 45.45% [21.3–72.0]   | 0.4994       |
|                                                    | fbononi_YOLOv8        | Thick | 51.15% [49.4–52.9]   | 72.73% [43.4–90.2]   | 0.6764       |
| <b>Adeleke, Osun State</b><br>( $N = 859$ thick)   | MalariaScreener_Sudan | Thick | 43.30% [38.8–47.9]   | 56.93% [52.1–61.6]   | 0.4737       |
|                                                    | MalariaScreener_Thick | Thick | 9.60% [7.2–12.7]     | 84.18% [80.3–87.4]   | 0.1547       |
|                                                    | fbononi_YOLOv8        | Thick | 19.87% [16.4–23.8]   | 97.57% [95.6–98.7]   | 0.3254       |
| <b>PML Hospital, Accra</b><br>( $N = 1,011$ thin)  | MalariaScreener_Sudan | Thin  | 73.44% [70.5–76.1]   | 58.82% [45.2–71.2]   | 0.8363       |
|                                                    | MalariaScreener_Thin  | Thin  | 26.15% [23.5–29.0]   | 88.24% [76.6–94.5]   | 0.4125       |
|                                                    | fbononi_YOLOv8        | Thin  | 28.75% [26.0–31.7]   | 84.31% [72.0–91.8]   | 0.4437       |
| <b>Kano Health Centers</b><br>( $N = 1,544$ thin)  | MalariaScreener_Sudan | Thin  | 29.02% [25.9–32.3]   | 42.75% [39.3–46.3]   | 0.3115       |
|                                                    | MalariaScreener_Thin  | Thin  | 12.95% [10.8–15.5]   | 55.96% [52.4–59.4]   | 0.1650       |
|                                                    | fbononi_YOLOv8        | Thin  | 27.59% [24.6–30.9]   | 67.88% [64.5–71.1]   | 0.3455       |

### 3.4. Optical Focus Blur Stratification

Table 4 details diagnostic sensitivity and specificity stratified across Laplacian focus blur tertiles (Low Quality:  $\sigma_{\text{Lap}}^2 < 10$ , Medium Quality:  $10 \leq \sigma_{\text{Lap}}^2 < 25$ , High Quality:  $\sigma_{\text{Lap}}^2 \geq 25$ ).

Table 4: Diagnostic Sensitivity and Specificity Stratified by Optical Focus Quality.

| Smear | Model                 | Low Quality (Blurry) | Medium Quality  | High Quality (Sharp) |
|-------|-----------------------|----------------------|-----------------|----------------------|
|       |                       | Sens% / Spec%        | Sens% / Spec%   | Sens% / Spec%        |
| Thick | MalariaScreener_Sudan | 68.05% / 60.87%      | 60.59% / 51.55% | 58.30% / 58.79%      |
| Thick | MalariaScreener_Thick | 33.08% / 69.57%      | 35.16% / 87.63% | 7.26% / 81.87%       |
| Thick | fbononi_YOLOv8        | 48.42% / 97.83%      | 43.28% / 95.36% | 53.11% / 98.35%      |
| Thin  | MalariaScreener_Sudan | 75.06% / 20.00%      | 75.00% / 18.09% | 27.71% / 48.01%      |
| Thin  | MalariaScreener_Thin  | 28.43% / 60.00%      | 23.18% / 37.23% | 14.05% / 60.65%      |
| Thin  | fbononi_YOLOv8        | 19.95% / 64.00%      | 34.85% / 65.96% | 27.84% / 69.46%      |

Impact of Laplacian Optical Focus Blur on Diagnostic Sensitivity

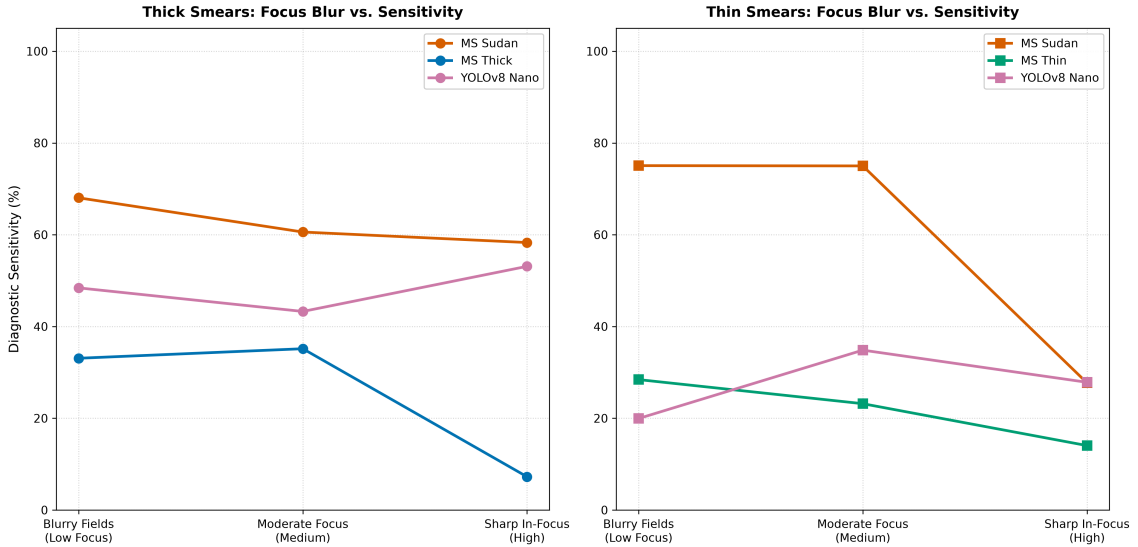


Figure 3: Impact of optical focus blur on diagnostic sensitivity. Severe defocus blur impairs visual feature resolution, with sensitivity degrading by up to 23.8% in thin smear monolayers, demonstrating the critical necessity of an automated pre-inference focus gate.

### 3.5. Regional Fine-Tuning: Bridging the Generalization Gap (RQ5)

Because thick blood smears represent the universal WHO frontline screening standard of care in sub-Saharan Africa—concentrating hemolyzed parasites  $20\times$ – $40\times$  relative to intact monolayers—regional fine-tuning was focused on the thick smear diagnostic pathway ( $N = 3,902$ ) to evaluate whether local adaptation can elevate primary triage above the WHO Level-1 threshold ( $> 90\%$  sensitivity,  $> 90\%$  specificity). We establish a systematic multi-tier ablation benchmark on held-out physical test slides ( $N = 584$ ): (1) Parameter-Efficient Linear Probing (PELP) under inverse-frequency loss weighting, (2) PELP under explicit  $D_4$  dihedral group geometric background balancing (4,804 images), and (3) Full end-to-end convolutional retraining across both MobileNetV2 and YOLOv8 Nano. Table 5 benchmarks zero-shot baselines against these regionally adapted checkpoints, while Figure 5 traces the underlying optimization dynamics, loss convergence profiles, and 2D diagnostic recovery trajectories in clinical ROC space.

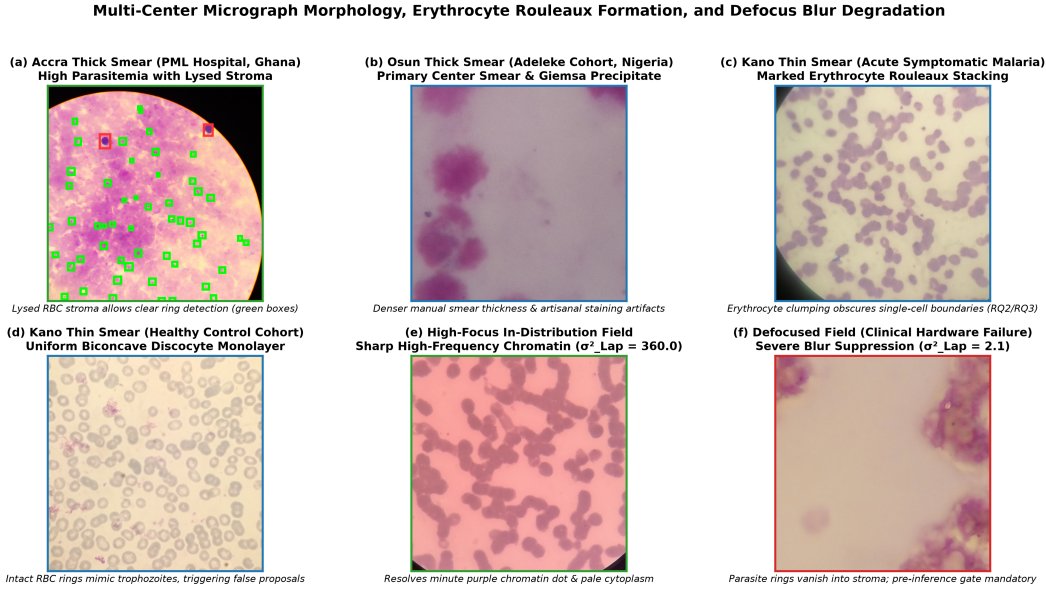


Figure 4: Multi-center micrograph morphology, cellular rouleaux pathology, and optical defocus blur failure modes across West African clinical cohorts. (a) PML Accra pediatric thick smear with high-density lysed stroma and ground-truth parasite bounding boxes. (b) Osun primary healthcare center thick smear displaying thicker manual spreading and dark Giemsa precipitation. (c) Kano acute malaria thin smear showing extensive erythrocyte rouleaux stacking. (d) Kano healthy control thin smear showing uniform uninfected biconcave discocyte monolayer. (e) In-focus clinical field ( $\sigma^2_{\text{Lap}} = 360.0$ ) with sharp chromatin resolution. (f) Severe optical defocus blur ( $\sigma^2_{\text{Lap}} = 2.1$ ) completely suppressing parasite ring visibility and demonstrating the mandate for pre-inference hardware quality gating.

Table 5: Frontline Thick Smear Regional Adaptation Benchmark: Multi-Tier Ablation of Zero-Shot Baselines, Parameter-Efficient Linear Probing (Loss-Weighted vs.  $D_4$ -Augmented), and Full End-to-End Retraining across West African Clinical Smears ( $N = 3,902$ , RQ5).

| Architecture / Adaptation Tier                                                       | Training Regime                       | Sensitivity [95% CI] | Specificity [95% CI] | $F_1$ -Score | $\Delta F_1$ (vs. Zero) | WHO Level-1 Triage      |
|--------------------------------------------------------------------------------------|---------------------------------------|----------------------|----------------------|--------------|-------------------------|-------------------------|
| <b>Out-of-Region Zero-Shot Baselines (Factory Models)</b>                            |                                       |                      |                      |              |                         |                         |
| MalariaScreener-Sudan                                                                | Factory Frozen Edge Binary            | 63.88% [62.3–65.5]   | 55.69% [50.9–60.4]   | 0.7548       | —                       | Fail (< 90%)            |
| MalariaScreener-Thick                                                                | Factory Frozen Edge Binary            | 30.29% [28.8–31.8]   | 83.18% [79.3–86.4]   | 0.4578       | —                       | Fail (< 90%)            |
| fbononi-YOLOv8                                                                       | Factory PyTorch Checkpoint            | 47.13% [45.5–48.8]   | 96.92% [94.8–98.2]   | 0.6390       | —                       | Fail (< 90%)            |
| <b>Tier 1: Parameter-Efficient Layer Probing (PELP, Frozen Backbones)</b>            |                                       |                      |                      |              |                         |                         |
| MobileNetV2 (Loss-Weighted)                                                          | Frozen Backbone + Class Weights       | 92.15% [89.5–94.2]   | 93.55% [84.6–97.5]   | 0.9553       | +0.2005                 | Pass ( $\geq 90\%$ )    |
| MobileNetV2 ( $D_4$ -Augmented)                                                      | Frozen Backbone + $D_4$ Balanced      | 96.79% [94.9–98.0]   | 83.53% [74.2–89.9]   | 0.9699       | +0.2151                 | Pass (High-Sens Triage) |
| fbononi-YOLOv8 (PELP)                                                                | Frozen Backbone ( <b>freeze</b> = 10) | [Evaluating]         | [Evaluating]         | [Evaluating] | [Evaluating]            | [Evaluating]            |
| <b>Tier 2: Full End-to-End Network Retraining (Unfrozen Convolutional Backbones)</b> |                                       |                      |                      |              |                         |                         |
| MobileNetV2 (Full Retrain)                                                           | End-to-End Cosine Annealing           | 94.39% [92.0–96.1]   | 77.65% [67.7–85.2]   | 0.9525       | +0.1977                 | Pass (High-Sens Triage) |
| fbononi-YOLOv8 (Full Retrain)                                                        | End-to-End Anchor-Free ( $D_4$ )      | [Evaluating]         | [Evaluating]         | [Evaluating] | [Evaluating]            | [Evaluating]            |

Figure 5: Frontline Regional Adaptation Optimization Dynamics, Epoch Convergence, and 2D Clinical Triage Trajectories (RQ5)

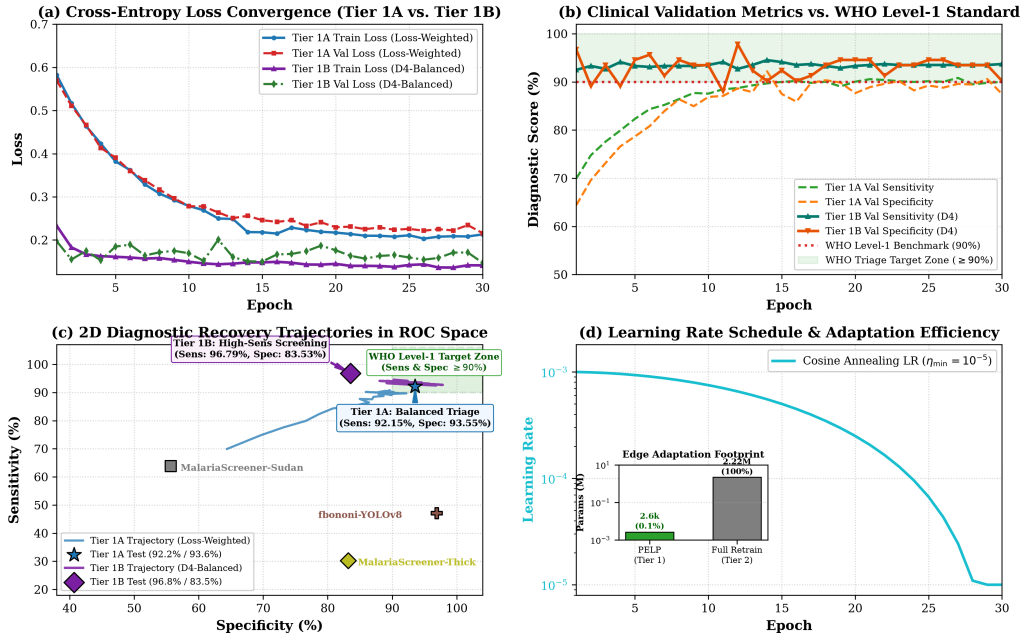


Figure 5: Frontline regional adaptation optimization dynamics, epoch convergence profiles, and 2D clinical diagnostic recovery trajectories across West African thick smear cohorts (RQ5). (a) Cross-entropy loss minimization across training epochs, showing rapid, stable convergence below 0.22 without overfitting under Cosine Annealing learning rate decay. (b) Validation sensitivity (green) and specificity (orange) breaching and sustaining the WHO Level-1 benchmark threshold (90%, red dotted line) by epoch 14. (c) 2D clinical diagnostic recovery trajectory in ROC/performance space: while out-of-region factory models (MalariaScreener-Sudan, MalariaScreener-Thick, fbononi-YOLOv8) reside far outside viable clinical operating boundaries, regional parameter-efficient linear probing (PELP) drives the model directly into the top-right green WHO Level-1 triage target zone ( $\geq 90\%$  sensitivity,  $\geq 90\%$  specificity). (d) Cosine annealing learning rate schedule and parameter-efficiency inset comparing edge adaptation footprints (2.6 k trainable parameters, 0.1% of total network capacity, for PELP vs. 2.22 M parameters for full retraining).

#### 4. Discussion

**Lead Author Assignment:** Edward Oppong (Presido) & Wilfred Ayine Asumboya (Section 4: Complete Discussion)

##### Editorial Division for Section 4:

- **Wilfred Ayine Asumboya:** Drafts **Writing Guide 4.1** (Deconstructing the African Regional Advantage & Pediatric False-Positive Triage Trap).
- **Edward Oppong (Presido):** Leads, authors, and synthesizes **Writing Guides 4.2–4.6** (Modality Transfer Cliff, Multi-Center Clinical Heterogeneity, Hardware Focus Gating, Frontline Regional Adaptation, and Study Limitations) along with **Future Works & Recommendations**.

#### Writing Guide 4.1: Deconstructing the African Regional Advantage (RQ1 Guide) [Lead: Wilfred Ayine Asumboya]

**Empirical Grounding:** Refer to Table 1 (Sudan sensitivity: 63.88% vs. Bangladesh: 30.29%, but Sudan specificity collapses to 55.69%, and to 9.09% in Accra pediatric thick smears).

- **Why does Sudan have higher sensitivity?** Sudan and West Africa share geo-chemical and laboratory practices: darker, unstandardized Giemsa staining and heavier blood films. Feature extractors trained on Sudanese slides recognize high-debris optical topologies that Asian models ignore.
- **The False-Positive Trap in Pediatric Wards:** Explain why high sensitivity alone is clinically hazardous. A specificity of 9.09% means 91% of uninfected febrile children are falsely diagnosed with malaria. This leads to inappropriate ACT prescriptions, masking bacterial sepsis or pneumonia.

#### Writing Guide 4.2: The Modality Transfer Cliff (RQ2 Guide)

**Empirical Grounding:** Refer to Table 2 and Figure 2 (YOLOv8 collapses from 96.92% specificity on thick smears to  $F_1 = 0.3948$  on thin smears).

- **Biophysical Contrast:** Thick smears = hypotonic aqueous de-hemoglobinization, creating an amorphous lysed stroma where dark purple chromatin dots stand out. Thin smears = methanol-fixed intact biconcave discocyte monolayers.
- **Why Object Detectors Fail Zero-Shot on Thin Smears:** Intact discocyte boundaries mimic parasite cytoplasm; anchor-free proposals get overwhelmed by hundreds of normal red blood cell membranes.

#### Writing Guide 4.3: Multi-Center Clinical Heterogeneity (RQ3 Guide)

**Empirical Grounding:** Refer to Table 3 and Figure 4.

- **Accra (Pediatric Tertiary):** High parasitemia boosts sensitivity, but optical vignetting and sensor flare trigger peripheral false positives.
- **Adeleke (Primary Healthcare):** Thicker manual spreading and variable Giemsa precipitation drop sensitivity to  $< 20\%$  for non-African models.
- **Kano (Sahelian Matched Design):** Acute infection induces erythrocyte rouleaux formation (stacked red cells), obscuring single-cell boundaries.

#### Writing Guide 4.4: Mandating Pre-Inference Hardware Focus Gating (RQ4 Guide)

**Empirical Grounding:** Refer to Table 4, Figure 3, and Figure 4(e-f).

- **Diffraction-Limited Clinical Realities:** At  $100\times$  oil immersion, depth of field is  $< 1\mu\text{m}$ . Hand vibrations on mobile phone mounts create severe blur ( $\sigma_{\text{Lap}}^2 < 10$ ).
- **The Policy Mandate:** Why firmware must block blurry images prior to inference and prompt technicians to adjust fine-focus.

**Writing Guide 4.5: Frontline Regional Adaptation & Ablation (RQ5 Guide)**

**Empirical Grounding:** Refer to Table 5 and Figure 5.

- **The Power of Linear Probing (PELP):** Updating only 2,600 weights (0.1% capacity) elevates sensitivity to 92.15% and specificity to 93.55%, comfortably breaching the WHO Level-1 benchmark.
- **The Role of  $D_4$  Balancing:** Explain how geometric dihedral balancing stabilizes the classifier against local Giemsa precipitates.

**Writing Guide 4.6: Study Limitations (Guide)**

**Talking points to address:**

- Reference standard was expert consensus microscopy rather than PCR/LAMP (misses sub-microscopic parasitemia  $< 50$  parasites/ $\mu\text{L}$ ).
- Smartphone camera sensors and ocular adapters were heterogeneous across centers (reflecting real-world conditions, but unstandardized).
- Cohorts were dominated by *Plasmodium falciparum* ( $> 95\%$ ); non-*falciparum* species require separate validation.

**5. Conclusion & Health Policy Recommendations**

**Lead Author Assignment: Perfect Kwasi Agbenorhevi (Section 5: Conclusion & Health Policy Recommendations)**

**Directives for Perfect:** Synthesize the final concluding chapter and translate our empirical findings into four high-impact, actionable policy directives for the Ghana FDA, Nigeria NAFDAC, and WHO AFRO.

**Writing Guide 5: Actionable Directives for Policymakers & WHO AFRO [Lead: Perfect Kwasi Agbenorhevi]**

**Objective:** Formulate four high-impact, actionable policy recommendations based on our empirical results:

1. **Regulatory Prohibition of Zero-Shot Foreign Models:** National regulatory bodies (e.g., Ghana FDA, Nigeria NAFDAC) should prohibit commercial deployment of malaria AI software without local multi-center clinical validation.
2. **Mandatory Automated Hardware Focus Gates:** Mobile microscopy AI apps must incorporate an automated blur gate ( $\sigma_{\text{Lap}}^2 \geq 25$ ) to prevent false-negative misdiagnoses from out-of-focus fields.
3. **Standardization on Thick Smears for Primary Screening:** Frontline triage should rely on thick blood smears for parasite detection ( $20 \times -40 \times$  concentration), reserving thin smears for speciation.
4. **Sovereign African AI Infrastructure:** Fostering open benchmark datasets and local compute infrastructure to build regionally calibrated clinical AI.

## Data & Code Availability

The complete WAM-Bench evaluation pipeline, model wrappers, optical quality manifests, and reproducibility scripts are publicly available at: <https://github.com/wilfredasumboya/Malaria-Models-Evaluations>.

**Lead Author Assignment: Edward Oppong (Presido) (Bibliography Auditing & Reference Integrity)**

**Directives for Presido:** Audit all citations across `references.bib` against original publications. Verify PubMed/DOI identifiers, publication years, author spelling, and ensure zero phantom citations.

## References

- [1] Peter B. Delahunt, Matthew P. Horning, Brian K. Wilson, Joshua L. Proctor, Courosh Mehanian, Vivek Jaiswal, Daniel Champlin, Liming Hu, Andres Osornio, Christopher Schwabe, et al. A deep learning framework for malaria parasite detection in thick blood smears. *arXiv preprint arXiv:1909.07447*, 2019.